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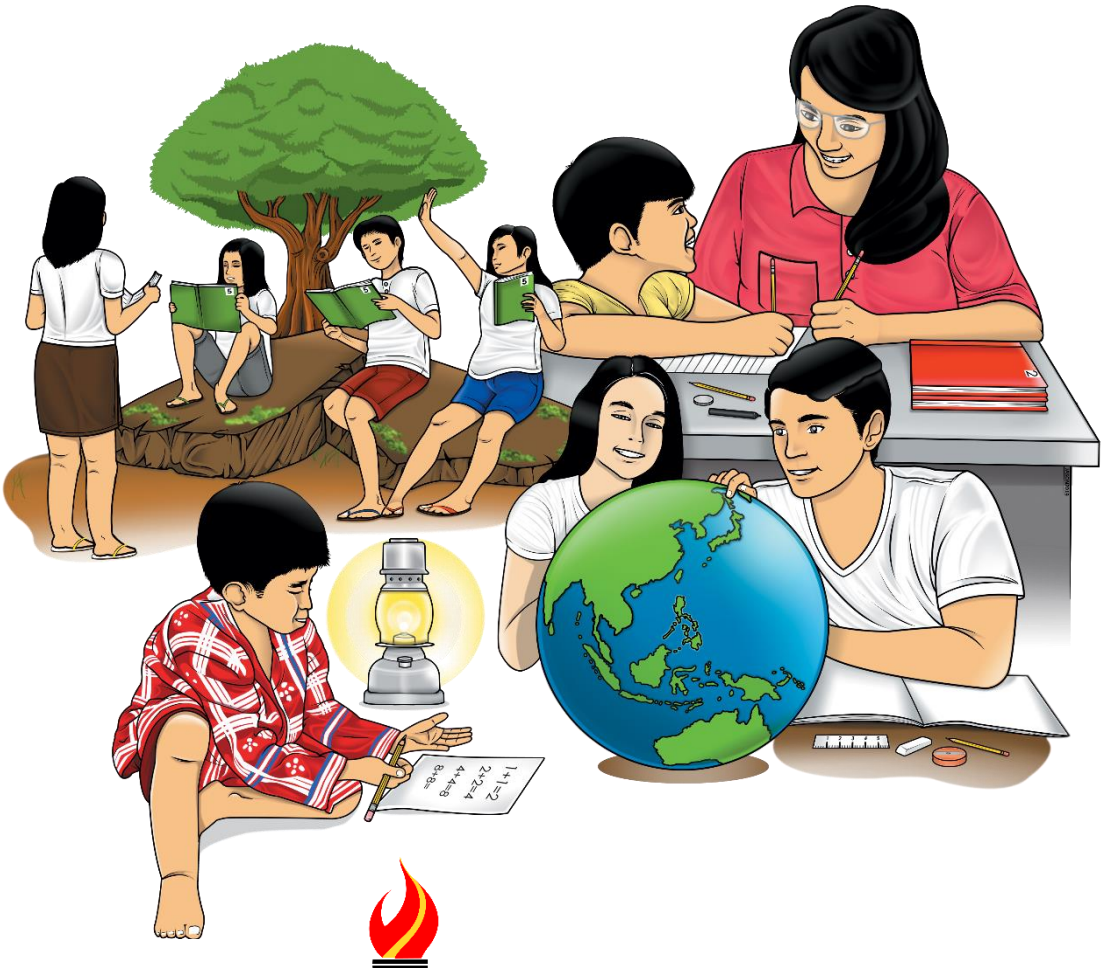
General Chemistry 1
Quarter 1



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General Chemistry 1 – Grade 12
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Quarter 1
First Edition, 2020

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Development Team of the Module

Authors:

Mary Rose Angelie S. Gatus, Sitero Francisco Memorial National High School
Rizalina E. Andamo, Malinta National High School
Bernadette Novilla, Valenzuela National High School
Jonathan F. Alvances, Valenzuela National High School
Mary Ann V. Dayrit, Valenzuela City School of Mathematics and Science
Virginia O. Bugay, Lingunan National High School
Jenalyn C. Guillermo, Paso De Blas National High School

Editor: Maria Lea Q. Prondo, Ed.D. - Education Program Supervisor - Science

Language Reviewer: Maria Celina J. Cloa, Canumay East Elementary School

Illustrator:

Layout Artist: Stephanie M. Salom, Pasolo Elementary School

Management Team:
Meliton P. Zurbano, Assistant Schools Division Superintendent (OIC-SDS)
Filmore R. Caballero, CID Chief
Jean A. Tropel, Division EPS in Charge of LRMS
Maria Lea O. Prondo, Ed.D., Education Program Supervisor - Science

Printed in the Philippines by:

Department of Education – National Capital Region

Division of Valenzuela
Office Address: Pio Valenzuela Street, Marulas, Valenzuela City
Telefax: 8-292-4340
E-mail Address: sdovalenzuela@deped.gov.ph

Table of Contents

Cover Page x

Development Team xx

Quarter 1: Week 1: Matter and Its Properties

Lesson 1. Properties of Matter and Formulas of Common Chemical Substances 1

Lesson 2. Consumer Products and Simple Separation Technique 5

Quarter 1: Week 2: Atoms, Molecules, and Ions

Lesson 3. Common Isotopes and Their Uses and Chemical Formulas 8

Lesson 4. Formulas and Names of Compounds and The Relationship of Percent Composition to Chemical Formula 12

Quarter 1: Week 3: Stoichiomentry

Lesson 5. Calculation of Molecular Formula Given the Molar Mass and Balanced Chemical Equation 16

Lesson 6. Mole or Mass Ratio 18

Quarter 1: Week 4: The Concept of Limiting Reagent

Lesson 7. Limiting and Excess Reagent 22

Quarter 1: Week 5: Gases

Lesson 8. Mass Relationship in a Chemical Reaction and Pressure 25

Lesson 9. Gas Laws and The Ideal Gas Equation 29

Lesson 10. Dalton's Law of Partial Pressure 34

Quarter 1: Week 6: Gas Stoichiometry

Lesson 11: Principles of Stoichiometry to determine the amounts (volume, number of moles, or mass) of gaseous reactants and products 38

Lesson 12: Relate the Rate of Gas Effusion with Molar Mass 40

References 43

Quarter 1 – Week 1:
Matter and Its Properties

What I Need to Know

This module was designed and written with you in mind. It is here to help you master matter and its properties. The scope of this module permits it to be used in many different learning situations. The language used recognizes the diverse vocabulary level of students. The lessons are arranged to follow the standard sequence of the course. But the order in which you read them can be changed to correspond with the textbook you are now using.

The module is divided into four lessons, namely:

- Lesson 1 – Properties of Matter
- Lesson 2 – Formulas of Common Chemical Substances
- Lesson 3 – Consumer Products
- Lesson 4 – Simple Separation Techniques

After going through this module, you are expected to:

1. Identify the use of properties of substances and how to separate them.
2. Appreciate the formulas of some common chemical substances.
3. List down consumer products that you can find in your home.
4. Identify these products based on their components, use, quality, and cost.
5. Illustrate the procedure of common separation techniques.

What I Know

Directions: Fill in the K-W-L Chart below to assess your prior knowledge and understanding of the topic.

What I Know?	What I Wonder?	What I Learned?

Lesson 1 — Properties of Matter and Formulas of Common Chemical Substances

Matter can be classified either as pure substance or mixtures, and they can be found around us. Every substance has a unique set of properties that allow us to differentiate between them. These properties are classified as physical properties and chemical properties.

At the end of this unit, you are expected to:

1. Use properties of matter to identify substances and to separate them. **STEM_GC11MPIa-b-5**
2. Recognize the formulas of common chemical substances. **STEM_GC11MPIa-b-9**

What's In

We are all surrounded by matter daily. Anything that we use, touch, eat, etc. is an example of matter. Matter can be defined or described as anything that takes up space, and it is composed of miniscule particles called atoms. It must display the two properties of mass and volume.

What's New

Trivia:

Do you know what the longest word is?

The longest word in English language is 189, 819 letters long. It is the chemical name of the molecule that confers passive elasticity on muscles. It takes three and a half to write down the word. Imagine having to write it down on a paper. Luckily, it has a shorter name, **Titin**.

What Is It

Properties of Matter

The different types of matter can be distinguished through two components: composition and properties. The composition of matter refers to the different components of matter along with their relative proportions. The properties of matter refer to the qualities/attributes that distinguish one sample of matter from another. These properties are generally grouped into two categories: physical or chemical.

Physical Properties are those that can be determined or measured without changing the composition or identity of the substance. These properties include *color, odor, density, melting point, boiling point, conductivity, and hardness.*

Chemical Properties tell us how the substance interacts with other substances and may include reaction with oxygen (oxidation), chlorine melts, etc. Determination of *chemical properties results* in the change of identity of a substance.

Matter is classified according to the property it exhibits. A property is something that distinguishes matter apart from other matters. These measurable properties are categorized into:

- **Intensive Property**
- are those properties that are not dependent on the amount of a substance and are used to identify a substance. Some properties, such as solubility, melting point, boiling point, and density are independent properties known as intensive properties.

- **Extensive Property**
- depend on the amount of a substance, such as mass and volume and are not useful in the identification of a substance.

Formula of Common Chemical Substances

We come across various chemical substances in our daily life. Some of the substance we use in our house comprise chemical substances. All these are bound up with atoms in a fixed proportion and also these are arranged in a certain way.

The **chemical formula** gives us a clear understanding regarding the chemical substances, like how many and what the chemical elements of the Periodic table consists of, along with the way they are arranged. Chemical formulas are used to express the composition of molecules and ionic compounds in terms of chemical symbols.

Chemical name is also important **because** it conveys some information about the structure of a compound. A common name is capable to identify a chemical compound in a particular set of circumstances. The name should indicate at least the chemical formula. It is important that each atom has a unique name.

What's More

Directions: A. Match column A with column B. Column **A** are chemical substances while column **B** are the formulas. Write only the letter of the correct answer.

COLUMN A	**COLUMN B**
_____ 1. Ammonia	a. H_2SO_4
_____ 2. Aspirin	b. H_2O_2
_____ 3. Citric acid	c. NH_3
_____ 4. Hydrogen peroxide	d. $C_9H_8O_7$
_____ 5. Muriatic acid	e. $C_6H_8O_7$

B. Complete the table below by checking the appropriate property of matter.

Property	Intensive Property	Extensive Property
1. Boiling Point		
2. Inertia		
3. Luster		
4. Length		
5. Density		

What I Have Learned

Directions: Complete the paragraph below.

Matter is characterized by physical and _____ properties. Physical properties pertain to those that are measured without _____ the composition of the substance. Chemical property, on the other hand, can be observed if a chemical _____ has occurred and thus there is a change in the _____ of the substance.

What I Can Do

Directions: Read and follow the procedures carefully.
Materials Needed: Candle, Matches
Procedure:

1. Observe the unlit candle. Touch or handle it to make your observations. Use all your senses-with the exception of taste. Write your observation in table 1 below.
2. Light the candle with a match. Be sure you place the lighted candle on a stable surface.
3. Observe the burning candle for 2 minutes. Write your observation in Table 1 below.
4. Extinguish the candle.

TABLE 1: OBSERVATIONS

Unlit Candle	Observation/s	Lighted Candle	Observation/s
Physical state		Physical state of the flame, wick	
Size		Size	
Shape		Color	
Odor		Temperature	
Color		Smoke	
		Brightness	

Guided Questions:

1. What are the physical properties of a candle?

_____

_____

2. What are the chemical properties of a candle?

_____

_____

3. How will you describe when the candle burns?

_____

_____

Assessment

Directions: A. Identify the following statement by arranging the jumbled letters in each number.

_____ 1. It depends on the nature of matter in a sample, not on the amount of matter present. (iinneetsv pprroeyt)

_____2. It is a characteristic that can be observed and measured without changing the identity and composition of the substance. (lacisyhp trprpeyo)

_____3. It depends on the amount of matter present. (teeeinsvx tyeorrpp)

_____4. It refers to the ability of a substance to undergo changes and to transform into a different substance. (cchemali rrppeoty)

_____5. It is a change in the composition of the substance. (lacimehc segnahc)

Directions: B. Below are examples of substances that we can find at home. Look for its formula.

_____ 6. toothpaste (sodium fluoride)

_____ 7. alcohol (ethanol)

_____ 8. detergent (sodium hydroxide)

_____ 9. baking soda

_____ 10. caffeine in coffee

Additional Activities

Directions: Classify the following as either *Physical Property* or *Chemical Property*.

_____ 1. Ozone in its pure form is blue in color.

_____ 2. Molybdenum steel used in rifle barrels is resistant to acids.

_____ 3. Mercury is a liquid metal.

_____ 4. The soil in some areas in Laguna is red.

_____ 5. Teflon is resistant to heat.

Lesson 2	Consumer Products and Simple Separation Technique

Any form of matter, either a pure substance or mixture is present all around us as it was mentioned in the earlier lessons. There are common substances that serve as essential products in our everyday living.

Matter can be classified either as pure substance or mixtures, and they can be found around us. A combination of two or more types of matter in varying amounts is called a mixture, and this can be separated using various simple separation techniques such as evaporation, distillation, filtration, and chromatography.

At the end of this unit, you are expected to:

1. Compare consumer products based on their components for use, safety, quality, and cost. **STEM_GC11MPIa-b-11**
2. Describe various simple separation techniques such as distillation, chromatography. **STEM_GC11MPIa-b-12**

What's In

At home**,** there are various consumer products that we used as food additives, beauty products, and cleaning materials. Try to look at the following pictures, can you name these products and identify their usage?

What's New

Directions: Write a short poem about the importance of consumer products in our everyday life.

_____

_____

_____

What Is It

Consumer Products

Materials for personal use such as soap, shampoo, and conditioner are found at home. Let us take soap as an example, it is used to get rid of the oily substances in our body. Soap has the chemical name sodium stearate and chemical formula of $C_{18}H_{35}O_2Na$. Food additives such as sodium chloride and monosodium glutamate or commonly known as table salt and umami seasoning respectively are substances that enhanced the flavor of our dishes.

We also have common commercial medicines that should be taken with proper prescription. Pain killer such as paracetamol which is sold in numerous brands are commonly found in the medicine kits at home. Cleaning products like hydrogen

chloride or also known as muriatic acid are chemical substances that should be handled with care because of its toxic effects that are harmful.

Simple Separation Techniques

Evaporation can be used as a method to separate a solid substance dissolved in liquid. This method requires heat where the liquid is evaporated, meaning it is converted from its liquid state to gaseous state. Once the liquid is completely evaporated, the solid substance is left behind.

Distillation is a technique that separates components of a liquid mixture by a process of heating and cooling. This technique employs the difference in volatility of the liquid components. This method is simple distillation, there are other forms like fractional distillation and steam distillation which uses much more complex instruments.

Filtration is a separation technique used to separate the components of a mixture containing an undissolved solid in a liquid. These undissolved particles are large enough to be filtered with a porous material. Another method which is related with filtration is **decantation**. In this process, the liquid is poured out leaving the solid particles at the bottom of the container. However, it is difficult to separate the substances without using any filtering tool.

Chromatography is the separation of a mixture by passing it in solution or suspension or as a vapor through a medium in which the components move at different rates. Other ways of separating mixtures include magnetization, crystallization, and centrifugation.

What's More

Directions: Read and follow the procedures below.

1. Draw a set-up of the following separation techniques in the space provided below.
2. Make sure to include labels of the instruments you are using in the set-up and instructions to follow in performing the procedure.
3. Be creative, add designs and colors to your output.
4. You can look at the figures present above to visualize your set-up.

1. Evaporation	2. **Filtration**	3. **Simple Distillation**	4. **Paper Chromatography**

What I Have Learned

Directions: Complete the statement. In this part you will write the things that you have learned in this lesson.

Mixture is a combination of two or more _____ and can be separated through various separation techniques. Evaporation separates a solid substance _____ in liquid. This method requires _____ where the liquid is evaporated

leaving the solid substance. _____ separates the components of a mixture containing an _____ solid in a liquid using a _____ material. Distillation is a technique that separates components of a liquid mixture by a process of _____ and cooling. This technique employs the difference in _____ of the liquid components. _____ is the separation of a mixture by passing it in solution or suspension. It can be used to separate mixtures that can be colored like _____ in plants.

What I Can Do

Directions: Read and answer the question below.

Why is it important to check the label as well as the contents and expiration date of products before purchasing them?

_____

_____

_____

Assessment

Directions: Determine the appropriate method to be used in separating components of the following substances.

_____ 1. Pigments present in plants
_____ 2. Salt solution
_____ 3. Chalk immersed in water
_____ 4. Dye
_____ 5. Crude oil
_____ 6. Sand & gravel
_____ 7. Alcoholic beverages
_____ 8. Brewed coffee
_____ 9. Mixture of iron fillings & wood chips
_____ 10. Solid components present in blood

Additional Activities

Directions: 1. Roam around your house (i.e. kitchen, bathroom, garage).
2. Look for 5 common products that your family usually use.
3. Fill in the table below.

Name of Product	Main Components/ Ingredients of the Product	Purpose/ Use	Safety (Is it safe to use?)	Quality (Is it effective?)	Cost/ Price
1.					
2.					
3.					
4.					
5.					

Quarter 1 – Week 2: Atoms, Molecules, and Ions

What I Need to Know

This module was designed and written with you in mind. It is here to help you master atoms, molecules, and ions. The scope of this module permits it to be used in many different learning situations. The language used recognizes the diverse vocabulary level of students. The lessons are arranged to follow the standard

sequence of the course. But the order in which you read them can be changed to correspond with the textbook you are now using.

The module is divided into four lessons, namely:

- Lesson 1 – Common Isotopes and Their Uses
- Lesson 2 – Chemical Formulas
- Lesson 3 – Formulas and Names of Compounds
- Lesson 4 – The Relationship of Percent Composition to Chemical Formula

After going through this module, you are expected to:

1. Recognize common isotopes and their uses.
2. Identify compounds using chemical formulas, structural formulas, and models.
3. Write names of inorganic compounds given their chemical formulas.
4. Explain the procedure used to calculate the percent composition by mass of a compound.

What I Know

Directions: Fill in the K-W-L Chart below to assess your prior knowledge and understanding of the topic.

What I Know?	What I Wonder?	What I Learned?

Lesson 3 — Common Isotopes and Their Uses and Chemical Formulas

We learned in the atoms and elements sections that each element has its own unique atom which is made up of a specific number of protons. The number of protons determines the atomic number of the element. Each atom also has the same number of electrons as protons.

At the end of this unit, you are expected to:

1. Recognize common isotopes and their uses. **STEM_GC11AMIc-e-19**
2. Represent compounds using chemical formulas, structural formulas, and models. **STEM_GC11AMIc-e-21**

What's In

Isotope, one of two or more species of atoms of a chemical element with the same atomic number and position in the periodic table and nearly identical chemical behavior but with different atomic masses and physical properties. Every chemical element has one or more isotopes.

What's New

Do you know what is an isotope?

Isotopes are atoms that have the same atomic number, but different mass numbers. Isotopes are denoted the same way as nuclides, but they are often symbolized only with the mass numbers because isotopes of the same element have

the same atomic number. Carbon, for example, has two naturally occurring isotopes, ${}^{12}_{6}C$ and ${}^{13}_{6}C$. Because both isotopes have 6 protons, they are often written as ^{12}C and ^{13}C. ^{12}C has 6 neutrons, and ^{13}C has 7 neutrons.

What Is It

Uses of Isotopes in Everyday Life

There are two types of isotopes, namely the stable isotopes (non-radioactive) and the unstable isotopes (radioactive).

Stable isotopes are non-radioactive forms of atoms. Although they do not emit radiation, their unique properties enable them to be used in a broad variety of applications, including water and soil management, environmental studies, nutrition assessment studies and forensics. While unstable isotopes go through radioactive decay and emit radiation and they are known as radioisotopes.

Several uses of radioisotopes in daily life are shown in the table below.

Field	Radioisotope	Use/s
Medical	Cobalt-60 (gamma rays)	Kill cancer cells without surgery in patients (radiotherapy)
	Phosphorus-32 and Strontium-90 (beta rays)	For cancer detection and treatment, especially in eyes and skin
	Sodium-24	Injected in patient's body in small amount as tracer. A radioactive detector is then used to detect accumulation of sodium-24 and therefore detect tumors and blood clots before they become dangerous.
	Plutonium-238 (from nuclear battery)	Produced small electric shocks in the heart pacemaker
	Iron-59	For anemia diagnosis
	Technetium-99	For brain, thyroid, liver, bone marrow, lung, heart, and intestinal scanning; blood volume determination
	Iodine-131	For diagnosis and treatment of thyroid function
	Xenon-133	For lung imaging
	Gold-198	For liver disease diagnosis
Agricultural	Phosphorus-32	For fertilizer
	Carbon-14	To study the passage of carbon during photosynthesis in plants
Industrial	Sodium-24	To detect leakage of underground pipes; and to measure the wear out rate of engine in a vehicle
	Sodium-24 (beta rays)	To control the thickness of plastic, paper and metal sheets in factory
	Sodium-24 (gamma rays)	To detect whether cans or bottles are filled up to the required amount
Food Preservation	Cobalt-60 (gamma rays)	To kill bacteria in food, to make vegetables fresh and last longer

		without any change in quality, flavor, and texture; and to inhibit budding in potatoes.
Archaeology	Carbon-14	To study and estimate the age of ancient artifacts (radiocarbon dating)

Chemical Formula

A chemical formula is an expression that shows the elements in a compound and the relative proportions of those elements.

Molecular Formula

A molecular formula is a chemical formula of a molecular compound that shows the kinds and numbers of atoms present in a molecule of the compound. Ammonia is a compound of nitrogen and hydrogen as shown below:

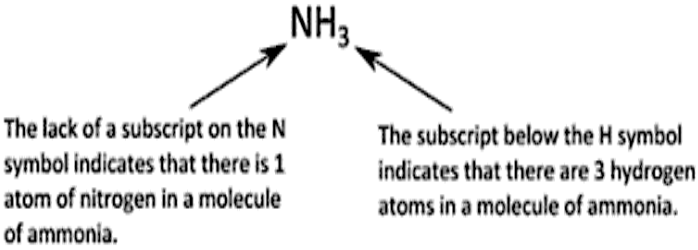


Figure 1: The molecular formula for ammonia
Source: https://chem.libretexts.org

Molecular formula for ammonia: NH_3. There is one atom of nitrogen and 3 atoms of hydrogen in a molecule of ammonia.

Note from the example that there are some standard rules to follow in writing molecular formula. The arrangements of the elements depend on the structure, so we will not concern ourselves with that point right now. The number of atoms of each kind is indicated by a subscript following the atom. If there is only one atom, no number is written. If there is more than one atom of a specific kind, the number is written as a subscript following the atom. We would not write N_3H for ammonia, because that would mean that there are three nitrogen atoms and one hydrogen atom in the molecule, which is incorrect.

Empirical Formula

An empirical formula is a formula that shows the elements in a compound in their lowest whole-number ratio. Glucose is an important simple sugar that cells use as their primary source of energy. Its molecular formula is $C_6H_{12}O_6$. Since each of the subscripts is divisible by 6, the empirical formula for glucose is CH_2O. When chemists analyze an unknown compound, often the first step is to determine its empirical formula.

- molecular formula: $C_6H_{12}O_6$
- empirical formula: CH_2O

There are many compounds whose molecular and empirical formulas are the same. If the molecular formula cannot be simplified into a smaller whole-number ratio, as in the case of H_2O or P_2O_5, then the empirical formula is also the molecular formula.

What's More

Directions: A. Complete the table below. Identify the missing information.

	Formula	Name	You have seen this substance before...	Element names	Number of atoms
1	CH_4	methane	Gas stoves for cooking	Carbon and hydrogen	1 x C 4 x H
2	H_2S	hydrogen sulfide	Rotten egg smell – flatulence		
3	O_2				
4			Fire extinguishers		1 x C 2 x O
5	CH_3COOH		Putting on chips		
6	$C_{12}H_{22}O_{11}$	sucrose	Table sugar		
7			Used to bleach hair		2 x H 2 x O
8	$C_8H_{10}N_4O_2$	caffeine			
9	$C_8H_9NO_2$	paracetamol			
10		copper			

What I Have Learned

Directions: Using available art materials at home, create a structural model of the compound isopropyl alcohol with the common name "rubbing alcohol" and a chemical formula $(CH_3)_2CHOH$.

What I Can Do

What do you think will happen if all the elements in the periodic table were non-reactive (stable isotopes)?

_____

_____

_____

_____

_____

Assessment

Directions: A. Use the word bank to match the isotope to its proper use/s. Then, write your answer on the space provided below after the number.

Carbon-14	Technicium-99	Plutonium-238
Sodium-24	Cobalt-60	Iodine-131

1. _____ is used for brain, thyroid, liver, bone marrow, lung, heart, and intestinal scanning; and blood volume determination.
2. _____ is used to study and estimate the age of ancient artifacts (radiocarbon dating).
3. _____ is used to produce small electric shocks in the heart pacemaker.
4. _____ is used to control the thickness of plastic, paper and metal sheets in factory chest cavity.
5. _____is used to kill cancer cells without surgery in patients (radiotherapy).

B. Complete the table of common names of certain chemical compounds below.

Common Name	Chemical Name	Chemical Formula
6. Baking soda		
7. Bleach (laundry)		
8. Bleaching powder		
9. Borax		
10. Caustic potash		

Additional Activities

Directions: Read and answer the question below.

What do you think will happen if all the elements in the periodic table were non-reactive (stable isotopes)?

_____

_____

_____

Lesson 4 Formulas and Names of Compounds and The Relationship of Percent Composition to Chemical Formula

Chemists use chemical formulas to express the composition of molecules and ionic compounds in terms of chemical symbols. By composition, we mean not only the elements present but also the ratios in which the atoms are combined.

At the end of this unit, you are expected to:

1. Name compounds given their formula and write formula given the name of the compound. **STEM_GC11AMIc-e-23**
2. Calculate the empirical formula from the percent composition of a compound. **STEM_GC11PCIf-32**

What's In

Chemical formulas are shorthand representations of substances, composed of symbols of elements written side by side to represent their compositions. A chemical formula tells what elements and numbers of atoms of those elements present in a compound.

What's New

Directions: Read and answer the question below.
What is percent composition?

_____

_____

_____

What Is It

Rules in Formula Writing

1. Formula for metals, non-metals and inert gases are also their symbol.
   Examples: sodium – Na chlorine - C1 neon - Ne
2. Common gases and the halogens are diatomic molecules, therefore, their formulas are written with a subscript of 2.
   Examples: hydrogen - H_2 nitrogen - N_2 chlorine – Cl_2
3. In writing the formula of a given compound, symbols for the positive element (cation) or radical is written first followed by the symbol of the negative element (anion) or radical Then, balance the positive and negative charges to make the compound neutral. When a radical is taken more than once, the radical is enclosed in parenthesis and the subscript is written outside the parenthesis (Crisscross Method).
   Examples: sodium chloride ($Na^+ Cl^-$) – NaC1
   aluminum sulfate ($Al^{+3} SO_4^{-2}$) – $Al_2(SO_4)_3$
4. Always given the simplest ratio of the atoms of ions that make the positive and negative charges add up to zero.
   Examples: ferrous oxide – Fe_2O_2 or FeO
   copper II sulfate – $Cu_2(SO_4)_2$ or $CuSO_4$
5. Formulas for most covalent molecules show the actual numbers of atoms in the molecule. Prefixes are commonly used like mono for one, di for two, tri for three, tetra for four, etc.
   Examples: carbon dioxide – CO_2 phosphorus pentachloride – PCl_5

Naming of Binary Compounds

In naming binary compounds, the name of the electropositive element or cation is written first followed by name of the electronegative element or anion, ending in -ide.

1. Binary Compounds of Metals and Non-metals: metallic element comes first followed by a non-metallic element.
   Examples: Na_2S – sodium sulfide NaC1 – sodium chloride
2. Binary Compounds with Variable Oxidation Number: two systems are used, <u>Stock System</u>: Metallic element is written first followed by the oxidation number written in Roman numeral and enclosed in parenthesis, followed by the name of the non-metal.
   Examples: CuC1 – copper (I) chloride $CuC1_2$ – copper (II) chloride

Old/Classical System: The suffix –ous is added to the Latin name of the cation with lower oxidation state and –ic for that with higher oxidation state.

Examples: CuC1 – cuprous chloride CuC1$_2$ – cupric chloride

3. Binary Compounds Containing Two Nonmetals: Prefixes are used like mono to indicate one atom, di for two atoms, tri for three atoms, tetra for four atoms, penta for five atoms etc.

   Examples: P_2O_5 – Diphospous pentoxide N_2O_3 – Dinitrogen trioxide

4. Salts of Binary Acids: These are formed when the hydrogen of an acid is replaced by a metal ion (HBr + NaOH = NaBr + H_2O). Name of the cation is given first followed by the anion.

   Examples: NaBr – Sodium bromide CaC1$_2$ – Calcium chloride

Naming of Ternary Compounds

Ternary compounds are compounds consisting of three or more elements, an electropositive group or cation, metal or hydrogen and a polyatomic negative group or anion.

1. Ternary Oxy-Acids: These are aqueous solutions made up of hydrogen, oxygen, and other elements. To indicate different oxidation states of elements, polyatomic anion's suffix –ate is changed to -ic, while suffix –ite is changed to –ous, followed by the word acid.

   Examples: HC1O: H+ ClO_3 (chlorite) – Chlorous acid

   H_3PO_4: H+ PO_4 (phosphate) – Phosphoric acid

2. Salts of Ternary Acids: These are formed when the hydrogen of an acid is replaced by a metal or ammonium ion (HCl + NH_4OH = NH_4Cl + H_2O). Name of the cation is given first followed by a polyatomic ion with appropriate suffix–ite (lower oxygen) or –ate (higher oxygen).

   Examples: K_2SO_3 – potassium sulfite K_2SO_4 – potassium sulfate

3. Salts with More than One Positive Ion: Salts can be formed from an acid containing two or more hydrogen atom by replacing one hydrogen or both hydrogen atoms with different metals. Positive group is named first, then the salt ending is added.

   Examples: $KHSO_4$ – potassium hydrogen sulfate or potassium bisulfate

   $CaHSO_3$ – calcium hydrogen sulfite or calcium bisulfite

Percent Composition

Percent composition is the percent by mass of each element in a compound. To calculate the percentage composition by mass of a compound, the (total) molar mass of an element is divided by the molar mass of the compound and multiplied by 100.

$$\% \text{ composition by mass} = \frac{\text{molar mass of element X}}{\text{molar mass of compound}} \times 100$$

What's More

Directions: Following the guidelines in naming inorganic compounds, complete the following table.

Chemical Formula	Name of compound	Binary or Ternary?
1. Fe_2O_3		
2. $HClO_4$		
3. N_7S_3		
4. $MgNH_4PO_4$		
5. $NH_4C_2H_3O_2$		

What I Have Learned

Directions: Construct a diagram or an organizer to show the step-by-step procedure of calculating the percent composition of a compound. Describe your diagram/organizer.

What I Can Do

Directions: Solve for the following problems. Write your answer in another sheet of paper.

1. Calculate the percent composition of H_2O.
2. The chemical formula of glucose is $C_6H_{12}O_6$. Determine its percent composition.
3. Which element comprising Ca3(PO4)2 has the highest percentage by mass?

Assessment

Directions: Give the names of the following compounds.

_____	1. NH_4Br	_____	6. LiCN
_____	2. $Sn_3(PO_3)_4$	_____	7. $Ca(HSO_4)_2$
_____	3. $H_2C_2O_4$	_____	8. $FePO_4$
_____	4. HI	_____	9. CBr_4
_____	5. PH_3	_____	10. $Pb(NO_3)_2$

Additional Activities

Directions: Accomplish the table to indicate the information that can be obtained from the chemical formulas given.

Chemical formula	Element/s present	Ratio of atoms	Total number of atoms
1. H_2O (Water)			
2. NH_3 (Ammonia)			
3. NaCl (Sodium chloride)			
4. $Al_2(SO_4)_3$ (Aluminum sulfate)			
5. P_2Cl_5 (Diphosphorus pentchloride)			

Quarter 1 – Week 3: Stoichiometry

What I Need to Know

This module was designed and written with you in mind. It is here to help you master stoichiometry. The scope of this module permits it to be used in many different learning situations. The language used recognizes the diverse vocabulary level of students. The lessons are arranged to follow the standard sequence of the

course. But the order in which you read them can be changed to correspond with the textbook you are now using.

The module is divided into three lessons, namely:

- Lesson 1 – Calculation of Molecular Formula Given the Molar Mass
- Lesson 2 – Balanced Chemical Equation
- Lesson 3 – Mole or Mass Ratio

After going through this module, you are expected to:

1. Calculate molecular formula given molar mass.
2. Write and balance chemical equation.
3. Construct mole ratio from a reaction.
4. Calculate the number of moles of reactants or products using mole ratios.

What I Know

Directions: Fill in the K-W-L Chart below to assess your prior knowledge and understanding of the topic.

What I Know?	What I Wonder?	What I Learned?

Lesson 5 Calculation of Molecular Formula Given the Molar Mass and Balanced Chemical Equation

Stoichiometry is a section of chemistry that involves using relationships between reactants and/or products in a chemical reaction to determine desired quantitative data. At the end of this unit, you are expected to:

1. Calculate molecular formula given molar mass. **STEM_GC11PCIf-33**
2. Write and balanced chemical equations. **STEM_GC11PCIf-g-37**

What's In

A chemical equation is the symbolic representation of a chemical reaction in the form of symbols and formulae, wherein the reactant entities are given on the left-hand side and the product entities on the right-hand side. The coefficients next to the symbols and formulae of entities are the absolute values of the stoichiometric numbers. The first chemical equation was diagrammed by Jean Beguin in 1615.

What's New

Did you know that?

In Greek, *stoikhein* means element and metron means measure, so stoichiometry literally translated means the measure of elements. To use stoichiometry to run calculations about chemical reactions, it is important to first understand the relationships that exist between products and reactants and why they exist, which require understanding how to balance reactions.

What Is It

The molecular formula is a formula giving the number of atoms of each of the elements present in one molecule of a specific compound. It also indicates the exact number of atoms in the molecule. One way of solving the molecular formula of a given substance or compound is by knowing its molar mass and empirical formula mass.

Molecular formula = (molar mass)/ (mass of empirical formula)

The **molar mass** is the mass of a given chemical element or chemical compound (g) divided by the amount of substance (mol).

Utilize molar mass data to obtain the molecular formula from the empirical formula.

Writing and Balancing Chemical Equation

In a chemical reaction, a substance/s is/are converted to one or more new substances. Chemical reactions follow the law of conservation of mass. No atoms are created or destroyed; they are just rearranged. Chemists have a way of communicating chemical reactions. They represent chemical reactions through chemical equations.

1. Identify reactants and products and write their correct formulas. Put reactants on the left side and products on the right.

$$C_2H_6 + O_2 \rightarrow CO_2 + H_2O$$

2. Balance the equation by changing the coefficients of the reactants or products. Do not change the subscripts or the chemical formula.

$$C_2H_6 + 7/2\ O_2 \rightarrow 2CO_2 + 3\ H_2O$$

To use the smallest whole number coefficients, we multiply the equation by 2 to give:

$$2C_2H_6 + 7O_2 \rightarrow 4CO_2 + 6H_2O$$

3. Check to make sure that the number of each type of atom is the same on each side of the equation.

Reactants	Products
4 C	4 C
12 H	12 H
14 O	14 O

What's More

Directions: Given are the following examples of chemical reactions. Balance the chemical equation.

1. **Decomposition reaction** – a reactant breaks down into two or more products.

$$___N_2O_5 \rightarrow ___N_2O_4 + ___O_2$$

2. **Synthesis reaction** – two or more reactants form a single product.

$$___NO + O_2 \rightarrow 2NO_2$$

3. **Single displacement reaction** – one element replaces another in a compound.

$$Cu_{(s)} + ___AgNO_{3(aq)} \rightarrow Cu(NO_3)_{2(aq)} + 2Ag_{(s)}$$

4. **Double displacement** – two ionic compounds exchange ions

$$2KI_{(aq)} + Pb(NO_3)_{2(aq)} \rightarrow __KNO_{3(aq)} + PbI_{2(s)}$$

5. **Combustion reaction** – a hydrocarbon (a compound containing carbon and hydrogen) reacts with oxygen to form carbon dioxide and water.

$$2C_2H_6 + ___O_2 \rightarrow 4CO_2 + 6H_2O$$

What I Have Learned

Directions: Answer the following questions below.

A white powder is found to have an empirical formula of P_2O_5. The compound has a molar mass of 283.88g. What is the compound's molecular formula?

1. Find the (empirical) molar mass of the empirical formula.
   P = _____ = _____
   O = _____ = _____
   (empirical)molar mass = _____
2. Divide the given molar mass by empirical molar mass = given molar mass/empirical molar mass = _____/_____ = _____
3. Multiply subscripts. = _____(P_2O_5) = _____

What I Can Do

Directions: Complete the table by giving the molecular formula of the following.

Empirical formula	P_2O_5	C_5H_{11}	NO_2	$C_6H_{12}O_6$	cyclopropane
Molecular formula					

Assessment

Directions: Solve the problem below and write your solution in another sheet of paper.

Tryptophan- a chemical found in some food is believed to make you sleepy has an empirical formula of $C_{11}H_{12}N_2O_2$ and empirical formula mass of about 204.22 g/mol. Find the molecular formula if the formula mass is 204.25 g/mol.

Additional Activities

Directions: Look at the food labels of some snack food (5) like potato chips, peanuts, popcorn, etc. Determine the percentage of sodium in snack food. Then answer the following in tabulated form. Use another sheet of paper for your answers.

1. Identify your chosen snack food and brand.
2. Get the amount in grams of one serving of the snack food.
3. Get the amount of sodium in milligrams in one serving of the snack food.
4. Obtain the percentage of sodium by mass in one serving of snack food.

Lesson 6 Mole or Mass Ratio

In a chemical reaction, compounds react in a set ratio. If the ratio is unbalanced, there will be leftover reactant. To understand this, you need to be familiar with the molar ratio or mole ratio.

At the end of this unit, you are expected to:

1. Construct mole or mass ratios for a reaction to calculate the amount of reactant needed or amount of product formed in terms of moles or mass. **STEM_GC11MRIg-h-38**

What's In

The mass of a mole of substance is called the *molar mass* of that substance. The molar mass is used to convert grams of a substance to moles and is used often in chemistry. The molar mass of an element is found on the periodic table, and it is the element's atomic weight in grams/mole (g/mol). If the mass of a substance is known, the *number of moles* in the substance can be calculated.

What's New

What is a mole?

Mole also spelled *mol*, a standard scientific unit for measuring large quantities of very small entities such as atoms, molecules, or other specified particles. The mole designates an extremely large number of units, $6.02214076 \times 10^{23}$.

What Is It

A **mole ratio** is a conversion factor that relates the amounts in moles of any two substances in a chemical reaction. The numbers in a conversion factor come from the coefficients of the balanced chemical equation.

Steps on How to Construct the Mole Ratio

1. Balance the reaction $N_2 + H_2 \rightarrow NH_3$.

$$N_2 + 3H_2 \rightarrow 2NH_3$$

2. Construct as many fractions (mole ratio) as you can making the reactants and products together with their number of moles as numerator and denominator.

$$N_2 + 3H_2 \rightarrow 2NH_3$$

The six mole ratios that can be written from the reaction above are:

$$\frac{1 \text{ mol } N_2}{3 \text{ mol } H_2}, \quad \frac{1 \text{ mol } N_2}{2 \text{ mol } NH_3}, \quad \frac{3 \text{ mol } H_2}{1 \text{ mol } N_2}, \quad \frac{3 \text{ mol } H_2}{2 \text{ mol } NH_3}, \quad \frac{2 \text{ mol } NH_3}{1 \text{ mol } N_2}, \quad \frac{2 \text{ mol } NH_3}{3 \text{ mol } H_2}$$

Stoichiometry problems can be characterized by two things: *(1) the information given in the problem, and (2) the information that is to be solved for,* referred to as *the unknown* and *the mole ratio*. The given and the unknown may both be reactants, both be products, or one may be a reactant while the other is a product. The amounts of the substances can be expressed in moles which can be converted to mass.

Amount of given substances in moles X Mole ratio = Amount of unknown substance in moles

Sample Problem 1

How many moles of ammonia are produced if 4.20 moles of hydrogen are made to react with nitrogen?

$$N_{2(g)} + H_{2(g)} \rightarrow NH_{3(g)}$$

Solution: STEP 1: Balance the equation.

$N_{2(g)} + 3H_{2(g)} \rightarrow 2NH_{3(g)}$

STEP 2: List the known quantities and plan the problem.

Given: H_2 = 4.20 mol H_2

Unknown: mol of NH_3

Note: Choose the mole ratio that has the NH_3 in the numerator (given) and the H_2 in the denominator (unknown).

STEP 3: Solve.

$$4.20\ mol\ H_2\ X\ \frac{2\ mol\ NH_3}{3\ mol\ H_2} = 2.80\ mol\ NH_3$$

STEP 4: Result.

2.80 moles of NH_3 will be produced if 4.20 moles of H_2 is made to react.

What's More

Directions: Calculate the number of moles of reactants or products using the mole ratios.

1. ___$Fe_2O_{3(s)}$ + ___$CO_{(g)}$ → ___$Fe_{(s)}$ +___$CO_{2(g)}$
   a. Balance the equation
   b. Calculate the number of moles of CO that can react with 3.25 moles of Fe_2O_3
   c. Calculate the moles of Fe formed when 0.150 kg of Fe_2O_3 react

2. ___$C_6H_{12}O_{6(aq)}$ → ___$C_2H_5OH_{(aq)}$ + __ $CO_{2(g)}$
   a. Balance the equation
   b. How many moles of $C_6H_{12}O_6$ are needed to form 7.50 moles of C_2H_5OH?
   c. How many moles of CO_2 form when 7.50 moles of C_2H_5OH are produced?

3. ___Na_2SiO_4 + HF → SiF_6 + NaF + H_2O
   a. How many moles of NaF is formed when 0.500 mol of HF reacts with Na_2SiO_4?
   b. How many moles of Na_2SiO_4 can react with 0.800 moles of HF?

What I Have Learned

Directions: Fill in the words that will complete the paragraph below.

Stoichiometry is the relationship between _____ and _____ in a reaction. It is the _____ relation between the number of _____ (and therefore mass) of various products and reactants in a chemical reaction. Chemical reactions must be _____, or in other words, must have the same number of various atoms in the products as in the reactants.

What I Can Do

Directions: Solve for the following problem. Write your answer in another sheet of paper.

KO_2 is used in a closed system breathing apparatus. It removes carbon dioxide and water from exhaled air. The reaction for the removal of water is: $KO_2 + H_2O \rightarrow O_2 + KOH$. The KOH produced is used to remove carbon dioxide by the following reaction: $KOH + CO_2 \rightarrow KHCO_3$.

a. What mass of KO_2 produces 235 g of O_2?
b. What mass of CO_2 can be removed by 123 g of KO_2?

Assessment

Directions: Solve the problem below and write your solution in other sheet of paper.

How many moles of ammonia are produced if 4.20 moles of hydrogen are made to react with nitrogen?

$$N_2(g) + H_2(g) \rightarrow NH_3(g)$$

Additional Activities

Directions: Create a "mnemonic device" about mole or mass ratio. Use the box provided below.

Quarter 1 – Week 4: The Concept of Limiting Reagent

What I Need to Know

This module was designed and written with you in mind. It is here to help you master the concept of limiting reagent. The scope of this module permits it to be used in many different learning situations. The language used recognizes the diverse vocabulary level of students. The lessons are arranged to follow the standard sequence of the course. But the order in which you read them can be changed to correspond with the textbook you are now using.

After going through this module, you are expected to:

1. Identify the limiting reactant (limiting reagent) in each chemical reaction.
2. Calculate how much product will be produced from the limiting reagent.
3. Calculate how much reactant(s) remains when the reaction is complete.

What I Know

Directions: Fill in the K-W-L Chart below to assess your prior knowledge and understanding of the topic.

What I Know?	What I Wonder?	What I Learned?

Lesson 7	Limiting and Excess Reagent

Chemical reactions occur when chemical bonds between atoms are formed or broken. The substances that go into a chemical reaction are called the reactants, and the substances produced at the end of the reaction are known as the products.

At the end of this unit, you are expected to:

1. Explain the concept of limiting reagent in a chemical reaction; identify the excess reagent(s). **STEM_GC11MRIg-h-40**

What's In

When there is not enough of one reactant in a chemical reaction, the reaction stops abruptly. To figure out the amount of product produced, it must be determined reactant will limit the chemical reaction (the limiting reagent) and which reactant is in excess (the excess reagent). One way of finding the limiting reagent is by calculating the amount of product that can be formed by each reactant; the one that produces less product is the *limiting reagent*.

What's New

Consider the process it takes to make a ham sandwich. You need two slices of bread and one piece of ham to make each sandwich. How many complete sandwiches could you make if you had eighteen slices of bread and six slices of ham? Let's set this up like a chemical equation where the ham and bread are our reactants, and the sandwich is our product:

2 slices of bread + 1 piece of ham = 1 ham sandwich

Now we can use stoichiometry to determine the number of sandwiches we could make if we used all our reactants.

1. How many complete sandwiches could you make if you had eighteen slices of bread and six slices of ham?

   _____

2. Which reactant limits the number of sandwiches produced?

   _____

3. Which reactant is left over?

   _____

What Is It

The reactant used up first in the chemical reaction is called the **limiting reagent**. **Excess reagents** are reactants present in quantities greater than what is needed by the reaction. The key in recognizing which reactant is the limiting reagent is based on a mole-mass or mass-mass calculation: whichever reactant gives the lesser amount of product is the limiting reagent.

Steps in Determining the Limiting Reactant

1. Get the number of moles of each reactant. If the given amount for each reactant is already in moles, then proceed to step 2.
2. Calculate the number of moles of product using each reagent.
3. The one that yields the smallest number of moles of product is the limiting reagent and the other one is excess reactant.

What's More

Directions: Solve for the following problem by showing all the necessary solution.

1. Consider the reaction: $2Al_{(s)} + 3Cl_{2(g)} \rightarrow 2AlCl_{3(s)}$. A mixture of 1.50 mol of Al and 3.00 mol of Cl_2 is allowed to react.
   a. Which is the limiting reactant?
   _____
   b. How many moles of $AlCl_3$ are formed?
   _____
   c. How many moles of the excess reactant remain at the end of the reaction?
   _____

2. Take the reaction: $NH_3 + O_2 \rightarrow NO + H_2O$. In an experiment, 3.25 g of NH_3 are allowed to react with 3.50 g of O_2.
   a. Which reactant is the limiting reagent? _____
   b. How many grams of NO are formed? _____
   c. How much of the excess reactant remains after the reaction? _____

What I Have Learned

Directions: Answer the following questions below.

1. Differentiate limiting from excess reactant?
   _____
   _____
2. What are the steps in determining the limiting reactant?
   _____
   _____
3. How will you calculate the amount of excess reactant after the reaction?
   _____
   _____

What I Can Do

Directions: Solve the given problem on limiting and excess reactant. Show all the necessary solution.

The reaction: $4NH_3 + 5O_2 \rightarrow 4NO + 6H_2O$ is a one of the steps in the commercial process for converting ammonia (NH_3) to nitric acid (HNO_3). In a certain experiment, 4.50 g NH_3 is reacted with 7.50 g O_2.

a. Which is the limiting reactant?
_____

b. How many grams of H_2O are formed assuming 100% yield?
_____

c. How many grams of excess reactant remain after the reaction?
_____

Assessment

Directions: Read and analyze each question below. Encircle the letter of the best answer.

1. What is the limiting reactant?
   - A. The reactant that is used up last and prevents more product from being made.
   - B. That reactant that is never used up.
   - C. The reactant that is used up first and prevents more products from being made.
   - D. That reactant that is in excess and does not get used up in the reaction.
2. Say you take a reactant A and calculate the amount of moles of another reactant B required to use up all of A. How do you know which of two reactants is the limiting one?
   - A. You compare the calculated amount of B to the actual amount available. If more is required, then B is the limiting reactant. If less is required, then A is the limiting reactant.
   - B. You compare the calculated amount of B to the actual amount available. if more is required, then A is the limiting reactant. If less is required, then B is the limiting reactant.
   - C. If the calculated value of B is larger than the amount of A, then A is the limiting reactant.
   - D. If the calculated value of B is larger than the amount of A, then B is the limiting reactant.
3. The reaction $N_{2(g)} + 2O_{2(g)} \rightarrow N_2O_{4(g)}$ occurs in a closed container. If 8.0 moles $N_{2(g)}$ are made to react with 12.0 moles O_2, the limiting reagent and the theoretical yield of N_2O_4 are:
   - A. The limiting reagent is N2; the theoretical yield of N2O4 is 8.0 moles.
   - B. The limiting reagent is N2; the theoretical yield of N2O4 is 16.0 moles.
   - C. The limiting reagent is O2; the theoretical yield of N2O4 is 12.0 moles.
   - D. The limiting reagent is O2; the theoretical yield of N2O4 is 6.0 moles.
4. Calcium carbide, CaC_2, is an important preliminary chemical for industries producing synthetic fabrics and plastics. CaC_2 may be produced by heating calcium oxide with coke: $CaO + 3C \rightarrow CaC_2 + CO$. What is the amount of CaC_2 which can be produced from the reaction of excess calcium oxide and 10.2 g. of carbon? (Assume 100% efficiency of reaction for purposes of this problem.)
   - A. 18.1 g
   - B. 28.4 g
   - C. 20.8 g
   - D. 19.8 g
5. When 12 moles of O_2 reacts with 1.1 mole of $C_{10}H_8$, what is the limiting reactant?

$$C_{10}H_8 + 12O_2 \rightarrow 10CO_2 + 4H_2O$$

   - A. Oxygen
   - B. $C_{10}H_8$
   - C. water
   - D. carbon dioxide

Additional Activities

Directions: Read the problem and answer the following questions.

In an experiment, 5.00 g aluminum (Al) is heated with 25.0 g sulfur (S) to form aluminum sulfide. The equation of the reaction is: $2Al + 3S \rightarrow Al_2S_3$

a. Which is the limiting reactant? _____

b. How many grams of Aluminum sulfide will be formed? _____

c. How many grams of excess reactant will remain? _____

Quarter 1 – Week 5:
Gases

What I Need to Know

This module was designed and written with you in mind. It is here to help you master gases. The scope of this module permits it to be used in many different learning situations. The language used recognizes the diverse vocabulary level of students. The lessons are arranged to follow the standard sequence of the course. But the order in which you read them can be changed to correspond with the textbook you are now using.

After going through this module, you are expected to:

1. Determine mass relationship in a chemical reaction.
2. Define pressure and its common uses.
3. Use the gas laws to determine pressure, volume, or temperature of a gas under certain conditions of change.
4. Use the ideal gas equation to calculate pressure, volume, Temperature, or number of moles of a gas.
5. Use Dalton's law of partial pressure to relate mole fraction and partial pressure of gases in a mixture.

What I Know

Directions: Fill in the K-W-L Chart below to assess your prior knowledge and understanding of the topic.

What I Know?	What I Wonder?	What I Learned?

Lesson 8 — Mass Relationship in a Chemical Reaction and Pressure

A balanced chemical equation shows the adherence of a chemical reaction of the Law of conservation of mass, which states that the mass of the reactants must be equal to the mass of the products in a chemical reaction. This principle allows us to calculate the amount of a substance that a given amount of reactant can produce, or the amount of reactant necessary for a certain quantity of a product to be formed. Stoichiometry is a branch in chemistry that deals with mass relationships between reactants and products based on a balanced chemical equation.

At the end of this unit, you are expected to:

1. (LAB) Determine mass relationship in a chemical reaction.
   STEM_GC11MRIg-h-42
2. Define pressure and give the common units of pressure.
   STEM_GC11G-Ihi-43

What's In

The International System of Units (abbreviated SI from the French *Système International d'Unités*) is the basis of the metric system. The SI was established in 1960 and is based on the metre-kilogram-second system rather than the centimetre-gram-second system. The units are divided into two classes: base units and derived units.

What's New

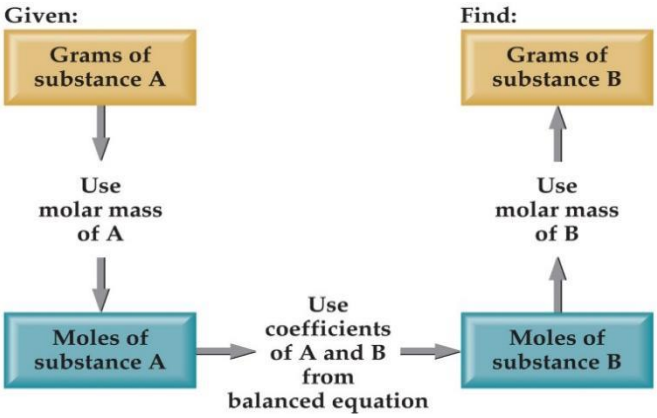
Can you guess how old this car is?

Maintaining adequate air pressure in an automobile tire has various advantages. Lowering the pressure makes the ride smoother and safer. The vehicle gets greater gas mileage and the tires last longer. The recommended tire pressure for that model of automobile (typically between 32 and 35 psi) is usually specified in the owner's manual or stamped inside the door. The tire is under pressure.

What Is It

Steps in Solving Stoichiometric Problems

1. Write a balanced equation.
2. Convert units of a given substance to moles.
3. Using the mole ratio, calculate the moles of substance yielded by the reaction.
4. Convert moles of wanted substance to desired units



Steps in Determining the Limiting Reactant

1. Get the number of moles of each reactant. If the given amount for each reactant is already in moles, then proceed to step 2.
2. Calculate the number of moles of product using each reagent.
3. The one that yields the smallest number of moles of product is the limiting reagent and the other one is excess reactant.

Calculating the Percent Yield

$$\text{percent yield} = \frac{\text{actual yield}}{\text{theoretical yield}} \times 100\%$$

Problem solving tip: *The first and most important step for any stoichiometric calculation—such as finding the limiting reagent or theoretical yield—is to start with a balanced reaction! Since our calculations use ratios based on the stoichiometric coefficients, our answers will be incorrect if the stoichiometric coefficients are not right.*

What's More

Directions: Carefully read each of the problems given below. Perform stoichiometric calculations based on the chemical reactions or chemical equations and then show your solution.

A. $2KClO_3 \longrightarrow 2KCl + 3O_2$

1. How many moles of O_2 will be formed from 1.65 moles of $KClO_3$?

2. How many moles of $KClO_3$ are needed to make 3.50 moles of KCl?

3. How many moles of KCl will be formed from 2.73 moles of $KClO_3$?

B. Acetylene gas, C_2H_2, undergoes combustion to form carbon dioxide and water when it is used in the oxyacetylene torch for welding. Balance the reaction and answer the following questions.

$$C_2H_{2(g)} + O_{2(g)} \longrightarrow CO_{2(g)} + H_2O_{(g)}$$

4. How many grams of water can be form if 113 g of acetylene is burned?

5. How many grams of acetylene react if 1.10 mol of CO_2 are produced?

What I Have Learned

Directions: Read and answer the following questions below.

1. On what law is stoichiometry based? Why is it essential to use balanced equations in solving stoichiometric problems?

   _____

   _____

2. Describe the steps involved in the mole method.

   _____

   _____

3. Define limiting reagent and excess reagent. What is the significance of the limiting reagent in predicting the amount of the product obtained in a reaction? Can there be a limiting reagent if only one reactant is present?

   _____

   _____

4. Why is yield of a reaction determined only by the amount of the limiting reagent?

   _____

   _____

5. Why is the actual yield of a reaction almost always smaller than the theoretical yield?

   _____

   _____

What I Can Do

Directions: Read and answer the question below.

Give an everyday example that illustrates the limiting reagent concept.

_____

_____

Assessment

Directions: Read and circle the letter of your answer.

1. Stoichiometry deals with
   - A. Combustion reactions
   - B. Rates of chemical reactions
   - C. Heat evolved or absorbed during chemical reactions
   - D. The study of amounts of materials consumed and products formed in chemical reactions
   - E. Activation energy of chemical reactions
2. Given the reaction $CH_4 + 2O_2 \rightarrow CO_2 + 2H_2O$, what amount of O_2 is needed to completely react with 14.0 moles CH_4?
   - A. 2.0 moles
   - B. 28.0 moles
   - C. 12.0 moles
   - D. 6.0 moles
   - E. 1.0 mole
3. How much of SnF_2 (stannous fluoride, active ingredient in toothpaste) in grams can be prepared from the reaction of 10.0 g SnO with excess HF according to the following reaction?

   $$SnO + 2HF \rightarrow SnF2 + H2O$$

   - A. 11.6 g
   - B. 10.0 g
   - C. 9.62 g
   - D. 26.0 g
   - E. 104.0 g

4. The reaction of 5.0 g hydrogen with 5.0 g carbon monoxide produced 4.5 g methanol. What is the percent yield for the reaction $2H_2 + CO \rightarrow CH_3OH$?
   A. 11%
   B. 79%
   C. 96%
   D. 24%
   E. 63%
5. The reaction of 5.0 g fluorine with excess chlorine produced 5.6 g ClF_3 in the reaction $Cl_2 + 3F_2 \rightarrow 2ClF_3$. What was the percent yield of the reaction?
   A. 58%
   B. 69%
   C. 76%
   D. 86%
   E. 92%

Additional Activities

Directions: Watch the animation videos of limiting reagent from the following sources:

• Limiting reagent [Vector animation]. Retrieved from McGraw Hill Education web site:
http://www.mhhe.com/physsci/chemistry/essentialchemistry/flash/limitr15.swf

• Limiting reactant [Vector animation]. Retrieved from North Carolina School of Medicine and Mathematics web site:
http://www.dlt.ncssm.edu/core/Chapter6-Stoichiometry/Chapter6-Animations/LimitingReactant.html

• Reactants, products and leftovers [Simulation]. Retrieved from PhEt Interactive Simulations web site: http://phet.colorado.edu/sims/html/reactants-products-and-leftovers/latest/reactants-products-and-leftovers_en.html

Gas Laws and The Ideal Gas Equation

Created in the early 17th century, the gas laws have been around to assist scientists in finding volumes, amount, pressures, and temperature when coming to matters of gas. The gas laws consist of three primary laws: Charles' Law, Boyle's Law and Avogadro's Law.

At the end of this unit, you are expected to:

1. Use the gas laws to determine pressure, volume, or temperature of a gas under certain conditions of change. **STEM_GC11G-Ihi-45**
2. Use the ideal gas equation to calculate pressure, volume, temperature, or number of moles of a gas. **STEM_GC11G-Ihi-46**

What's In

The relationship between pressure, temperature, volume, and amount of gas is discovered using the three fundamental gas laws. The volume of gas grows as the pressure decreases, according to Boyle's Law. According to Charles' Law, the volume of a gas increases as the temperature rises. Avogadro's Law states that as the volume of gas increases, so does the amount of gas. The combination of the three fundamental gas law is the ideal gas law.

What's New

So how do we "count" chemical entities involved in a chemical reaction?

Macroscopic observables such as mass, volume, temperature, and pressure are used to count chemical entities in a process. The condition of matter determines what we measure, but we must link a quantifiable quality to the quantity of particles. The ideal gas law (PV=n) will be used in the gas phase chapter.

What Is It

Boyle's Law

The relationship between the volume and pressure of gases at constant temperature was first stated by Robert Boyle during the 16th century. He performed an experiment wherein he trapped a fixed amount of air in the J- tube, he changed the pressure and controlled the temperature and then, he observed its effect to the volume of the air inside the J-tube. He found out that as the pressure is increased, the volume decreases. He finally concluded that **the volume of a fixed amount of gas is inversely proportional so its pressure at constant temperature.**

Let 's try to solve this problem. A sample of gas with a volume of 1.45 L exerts a pressure of 780 torr. What must the volume be if the pressure is to become 740 torr, at constant temperature and amount of gas?

Given:	**Find:**	**Formula:**	**Solution:**
V_1 = 1.45 L	V_2 =?	$P_1V_1 = P_2V_2$	$V_2 = \frac{(780\ \cancel{torr})(1.45L)}{740\ \cancel{torr}}$
P_1 = 780 torr		$V_2 = \frac{P_1V_1}{P_2}$	V_2= 1.53 L
P_2 = 740 torr			

Charle's Law

Charles' law, a statement that the **volume** occupied by a fixed amount of gas is **directly proportional** to its absolute temperature, if the pressure remains constant. The volume-temperature relationship in gases (k = V/T) was determined by and named after Jacques Charles.

Mathematically, Charles' Law can be expressed as **V α T at constant P** where **V** = volume and **T** = temperature expressed in Kelvin. *Note: Degree Celsius (°C) must be converted into Kelvin (K) because it is the basic unit for measuring temperature in the International Standards (SI). "It denotes the absolute temperature scale whereby 0K or absolute zero is defined as the temperature when molecules will have the lowest energy."*

Removing the proportionality symbol (α) and using the equality sign (=) the equation will be as follows: $\mathbf{V = kT}$ or $\boldsymbol{k = \frac{V}{T}}$. Thus, in a direct proportion, the quotient of the variable is constant.

If you are going to consider the initial and final conditions, you will arrive at the following equations: $\frac{V_1}{T_1} = \mathbf{k}$ and $\frac{V_2}{T_2} = \mathbf{k}$ whereas $\mathbf{V_1}$ is the initial volume and $\mathbf{V_2}$ is the final volume, and $\mathbf{T_1}$ is the initial temperature and $\mathbf{T_2}$ is the final temperature.

If the volume-temperature ratios are the same in the initial and final conditions, then we will arrive at this equation: $\frac{V_1}{T_1} = \frac{V_2}{T_2}$.

Sample Problem

An inflated balloon with a volume of 0.75 L at 30° C was placed inside the freezer where the temperature is -10°C. Find out what will happen to the volume of the balloon if the pressure remains constant. Support your answer with computation.

Initial Conditions	Final Conditions	
V_1 = 0.75 L	V_2 = ?	**Note:** *Convert °C into Kelvin (K) by adding 273 K* Solve for the final volume. **Note:** *Derive the formula to solve for the final volume.*
T_1 = 30 °C + 273 K = 303 K	T_2 = -10 °C + 273 K = 263 K	

$$V_2 = \frac{V_1 T_2}{T_1} = \frac{(0.75\ L)(263\ \cancel{K})}{303\ \cancel{K}} = \frac{197.25\ L}{303} = 0.65\ L$$

Avogadro's Law

When you inflate a tire or a balloon, its volume and pressure will increase because the amount of gas molecules increases as you pump. According to Avogadro's law (named after an Italian chemist Amedeo Avogadro), equal volumes of different gases at the same temperature and pressure contain the same number of molecules. Thus, one mole of any ideal gas at standard temperature and pressure (STP) occupies a constant volume of 22.4 L. It is important to remember that Avogadro's law applies only to substances in the gaseous state measured at the same temperature and pressure. Thus operationally, at constant pressure and temperature, the volume of a gas is directly proportional to the number of moles of a gas as shown in $\frac{V1}{n1} = \frac{V2}{n2}$. ($V_1$ = initial volume, V_2 = final volume, n_1 = initial amount of gas, n_2 = final amount of gas). In mathematical form, this equation is written as $\mathbf{V_1n_2 = V_2n_1}$.

Sample Problem

A 5.00 L of a gas is known to contain 0.965 mol. If the amount of gas is increased to 1.80 mol, what new volume will result?

Given: V_1= 5.00 L, n_1= 0.965 mol, n_2= 1.80 mol **Formula:** $V_1n_2 = V_2n_1$
Solution: V_2 = (5.00 L)(1.80 mol)/0.965 mol **Answer:** V2 = 9.33 L (V ↑ n ↑)

Ideal Gas Law

An ideal gas is defined as a hypothetical gaseous substance whose behavior is independent of attractive and repulsive forces and can be completely described by the ideal gas law. But there is no such thing as an ideal or perfect gas, but an ideal gas is a useful conceptual model that allows us to understand how gases respond to changing conditions. As we shall see, under many conditions, most real gases exhibit behavior that closely approximates that of an ideal gas. The ideal gas law can therefore be used to predict the behavior of real gases under most conditions.
Deriving the Ideal Gas Law.

Combining the Boyle's Law, Charles' Law, and Avogadro's Law it gives us $V \propto \frac{nT}{P}$. Which shows that the volume (V) of a gas is proportional to the numbers of moles (n) and the temperature (T) and inversely proportional to the pressure (P). Then, $V \propto$ constant($\frac{nT}{P}$). The proportionality Constant in Equation is called the gas constant, which is R = 0.0821 $\frac{L.atm}{K.mol}$. Rearranging gives us the Ideal Gas Equation: $V=\frac{RnT}{P}$ or

PV=nRT where: V=volume in Liters (L), P=pressure in Atmosphere (atm), T=temperature in Kelvin (K), n=number of moles (moles), and R=universal gas constant.

Sample Problem

Calculate the pressure exerted by a 0.25 mole sulfur hexafluoride in a steel vessel having a capacity of 1.25 L at 70.0℃.

Given: n = 0.25 mole, V = 1.25 L, T = 70.0 ℃ = 343.15 K, R = 0.0821 $\frac{L.atm}{K.mol}$
Find: P =?
Formula: PV = nRT **Note:** To find the formula for the pressure. $\frac{PV}{V} = \frac{nRT}{V}$
Therefore: P = $\frac{nRT}{V}$
Substitution: P = $\frac{0.25 \text{ mole } (0.0821 \frac{L.atm}{K.mol})(343.15K)}{1.25 \text{ L}}$ **Note:** Cancel same units.

Answer: P = 5.63 atm

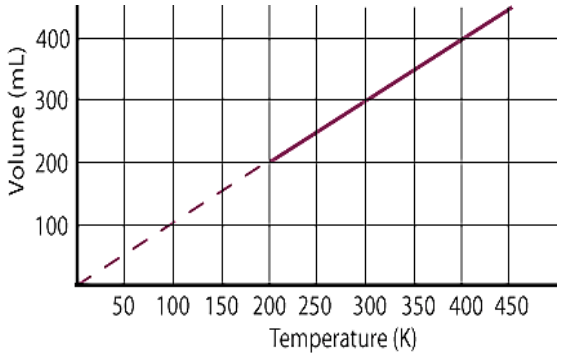
What's More

Directions: A. Put a check (/) on the blank if the given example is from the application of Boyle's Law and cross out (X) if not.
_____1. The scuba diver's activity.
_____2. On how the hot air balloon works.
_____3. The use of lung model in respiratory system.

B. Choose the letter of the best answer and write it on the space provided before the number.
_____4. What happens to a volume of a gas if the pressure decreases at constant temperature?
A. decreases B. increases C. stays the same D. unpredictable
_____5. When pressure on a gas goes down, what happens to its volume?
A. goes down B. rises C. rises, then falls D. stays the same

What I Have Learned

Directions: Analyze the given graph below and answer the guide questions.



Guided Questions:
1. What are the two variables involved in the graph?
_____
2. What happen to the volume of gas as temperature increases?
_____

3. What is the relationship between the volume and temperature of a gas at constant pressure?

_____

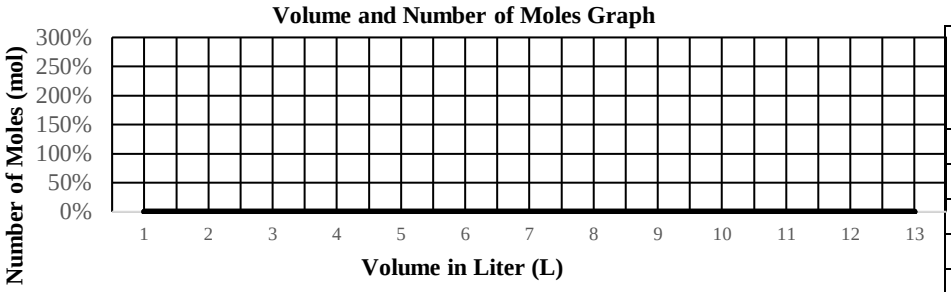
What I Can Do

Directions: Read and answer the question below.
What is the importance of Gas Laws in our everyday life?

_____
_____
_____
_____

Assessment

Directions: A. Show how volume and mole are related by plotting the values of volume and mole found in table on the right side to the Volume-Mole Graph below.



Volume in L	No. of moles in mol
2.50	0.50
5.00	1.0
7.50	1.5
10.00	2.0
12.50	2.5

Guided Questions:

1. Describe the volume and mole graph.

_____

B. Fill up the table on the right side by solving the unknown quantity, assuming temperature and pressure are constant.

V_1 in L	V_2 in L	n_1 in mol	n_2 in mol
2.50		0.68	1.36
	1.75	1.80	0.60
0.84	3.36	0.24	
7.50	15.0		2.40

C. Solve the following problem. Show your solution.

1. At what temperature will 0.654 moles of neon gas occupy 12.30 liters at 1.95 atmospheres?

_____

2. A 0.25 moles neon is at 256 mmHg and at temperature of 35°C. What is the volume? (1 atm = 760 mmHg) _____

Additional Activities

Directions: Solve the following problems and show your solution on the box provided below.

1. A scuba diver needs a diving tank to provide breathing gas while he is under water. How much pressure is needed for 6.00 liters of gas at 1.01 atmospheric pressure to be compressed in 3.00-liter cylinder?

2. A sample of nitrogen gas has a volume of 0.3 L at 750 mmHg. If the volume is changed to 0.1 L, what is the new pressure at constant temperature and amount of gas?

Lesson 10 Dalton's Law of Partial Pressure

Dalton's Law, or the Law of Partial Pressures, states that the total pressure exerted by a mixture of gases is equal to the sum of the partial pressures of the gases in the mixture.

At the end of this unit, you are expected to:

1. Use Dalton's law of partial pressures to relate mole fraction and partial pressure of gases in a mixture. **STEM_GC11DLIi-47**

What's In

We measure gas pressure in everyday life when we use a barometer to check the outside atmospheric pressure or a tire gauge to check the pressure in a bike tube. We're measuring a macroscopic physical attribute of a big number of gas molecules that aren't visible to the human eye when we accomplish this. The pressure we're measuring comes from the force of individual gas mols at the molecular level.

What's New

Who is John Dalton?

The behavior of mixtures of gases was described by John Dalton. He said that in an enclosed container, the total pressure exerted by the gases will be equal to the summation of the individual pressure exerted by the gas as if it were alone in a container.

What Is It

Dalton's law of partial pressures states that the total pressure of a mixture gases is the sum of the partial pressures of its components. This law is expressed in an equation as: $\boldsymbol{P_T = P_1 + P_2 + P_3 + \ldots + P_n}$

where: P_T is the total pressure of the mixture P_1, P_2, $P_{3,} \ldots . P_n$ are the partial pressure of all the component.

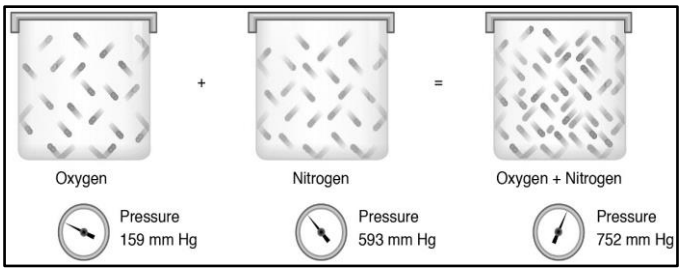


Figure 1. The partial pressure of a gas in a mixture is the same as the pressure of the gas in the container by itself. The sum of the partial pressures gives the total pressure of the gas mixture.

Sample Problem 1

About 2.75 moles of H_2 gas was mixed with 7.00 moles CO_2 gas. Determine the mole fraction of H_2 gas and CO_2 gas in the mixture.

Given: no. of moles of H_2 = 2.75 moles
no. of moles of CO_2 = 7.00 moles
Required: Mole fraction (X)
Solution: STEP 1. Find the total no. of moles using the formula

n_T = moles of H_2 + moles of CO_2
= 2.75 moles + 7.00 moles
n_T = 9.75 moles

STEP 2. Find the mole fraction of H_2 gas

$$X_{H2} = \frac{moles\ of\ H_2}{total\ no.of\ moles}$$
$$= \frac{2.75\ moles}{9.75\ moles}$$

Note: Cancel the same unit

$$X_{H2} = 0.28$$

STEP 3: Find the mole of CO_2 gas

$$X_{CO2} = \frac{moles\ of\ CO_2}{total\ no.\ of\ moles}$$
$$= \frac{7.00\ moles}{9.75\ moles}$$
$$X_{CO2} = 0.72$$

Sample Problem 2

A mixture of H_2 and He was placed in a 6.50 L sealed container and maintained at 40 °C. Calculate the total pressure if 2.55 moles each of H_2 and He were mixed.

Note: In this problem you will be using the formula of ideal gas law

$$PV = n\ RT$$

where: P is pressure, V is volume, T is temperature in K (273 K), n is amount of gas in moles, R is gas constant (0.0821 L-atm/mol K)

Given: no. of moles of H_2 = 2.55 moles
no. of moles of He = 2.55 moles
V = 6.50 L
T = 40.0 °C + 273K = 313K
Required: Total pressure (P_T)
Solution: STEP 1. The ideal gas equation for each mixture is given as:

$$P_{H2} = \frac{nH_2\ RT}{V}$$
$$P_{He} = \frac{n_{He}\ RT}{V}$$

STEP 2. The total pressure is given by**:**

$$P_T = P_{H2} + P_{He}$$

STEP 3. Substitute the partial pressure of each gas as:

$$P_T = \frac{n_{H2}\ \ RT}{V} + \frac{n_{He}\ RT}{V}$$

STEP 4. Combine similar terms

$$P_T = n_{H2} + n_{He}\ [\frac{RT}{V}]$$

STEP 5. Substitute the given values:

$$P_T = 2.55\ moles + 2.55\ moles\ [\frac{0.0821\frac{L-atm}{mol\ K\ (313K)}}{6.50L}]$$
$$P_T = 5.1\ [\frac{25.70\ atm}{6.50}]$$
$$= \frac{131.07\ atm}{6.50}$$

Note: cancel the same unit

$$P_T = 20.17\ atm$$

What's More

Directions: Read and follow the procedures below.

Materials: Worksheet, Scientific Calculator, Pen, and Pad paper
Procedure: Read and understand carefully the procedure. Analyze the figure below.

1 mole of gas X 20 ◦C = 2.0 atm	✚	1 mole of gas Y 20 ◦C = 2.0 atm	⇨	1mol gas X + 1mol gas Y 20 ◦C = + = 2.0 + 2.0 = 4.0 atm = 40 L

1. What information can you obtained using the figure?
   A. In a mixture of gases, where no reaction takes place between the components, the total pressure is the sum of the partial pressures of the gas components.
   B. A mixture of gases has a reaction between the components.
   C. At constant temperature, the pressure of a fixed quantity of a gas varies inversely with the volume.
   D. At constant pressure, the volume of a fixed quantity of a gas is directly proportional to the absolute temperature.
2. Write the mathematical formula of Dalton's law of partial pressure.

   _____

   _____
3. What is the relationship between the amount of gas in a mixture and its partial pressure?

   _____

   _____

What I Have Learned

Directions: Complete the paragraph below by filling in the blanks with the correct word. Choose your answer from the box below.

apply	formulas	steps	systematic

Based on what I have learned in this lesson it is important to be more _____ in solving problems regarding Dalton's law of partial pressure and know the _____ and _____ to be used.

What I Can Do

Directions: Read and answer the question below.
When is the concept of pressure evident in our daily life activities?

_____

_____

_____

Assessment

Directions: Solve the problem below. Use another sheet of paper for your answer.
A mixture of N_2 and H_2 gases to be used for ammonia production exerts a pressure of 2.5 atm. If the mole fraction of N_2 in the mixture is 0.67, What are the individual pressures exerted by each gas in a mixture?

Additional Activities

Directions: Choose the best answer from the box as described by each statement below.

0.0821 L-atm/mol K	Kelvin (K)	Partial pressure
Mole fraction	Law of partial pressure	

_____1. The unit of temperature used in the computation involving gas laws.
_____2. It states that the total pressure exerted by a mixture of gases is the sum of the partial pressures exerted by the individual gases if it were alone in the container.
_____3. It refers to the number of moles of a component of a mixture divided by the total number of moles in the mixture.
_____4. It refers to the pressure exerted by one gas in a mixture of gases.
_____5. The gas constant, R in using ideal gas law.

Quarter 1 – Week 6: Gas Stoichiometry

What I Need to Know

This module was designed and written with you in mind. It is here to help you master gas stoichiometry. The scope of this module permits it to be used in many different learning situations. The language used recognizes the diverse vocabulary level of students. The lessons are arranged to follow the standard sequence of the course. But the order in which you read them can be changed to correspond with the textbook you are now using.

After going through this module, you are expected to:

1. Apply the principles of stoichiometry to determine the amounts (volume, number of moles, or mass) of gaseous reactants and products.
2. Relate the rate of gas effusion with molar mass.

What I Know

Directions: Fill in the K-W-L Chart below to assess your prior knowledge and understanding of the topic.

What I Know?	What I Wonder?	What I Learned?

Lesson 11	Principles of Stoichiometry to determine the amounts (volume, number of moles, or mass) of gaseous reactants and products

Stoichiometry is related to gas. It is the quantitative relationship between reactants and products in a balanced chemical equation.

At the end of this unit, you are expected to:

1. Apply the principles of stoichiometry to determine the amounts (volume, number of moles, or mass) of gaseous reactants and products. **STEM_GC11GSIi-j-48**

What's In

Chemical reactions between gaseous materials are quite similar to reactions between solids and liquids, except the Ideal Gas Law (PV=nRT) can now be included in the calculations. If a chemical reaction is reversible (such as the decay and formation of dinitrogen tetraoxide), then Dalton's Law of Partial Pressure may be used to determine the moles of reactants and products at which the reaction ceases (and subsequently, the temperature, pressure and volume of each gas can be determined as well).

What's New

How is stoichiometry related to gas?

Stoichiometry will relate not only to the number of products and reactants in grams or moles but also to the behavior describing the number of gases such as such as volume, temperature, and pressure.

What Is It

The calculations of gas stoichiometry involve mass relationships in chemical reactions where a gas is either a reactant or a product. If the number of moles (n) of a gaseous reactant or product is known, the volume of the gas is calculated directly using the ideal gas equation. The gas law helps you determine the effect of temperature, pressure, and volume or the number of moles of a gas.

The requirement of any stoichiometric problem is to convert moles of A to moles of B. If A and/or B are solids or liquids, use the mass or molar mass to get the moles. If A and/or B are gases, use the Ideal Gas Law to get the moles.

Sample Problem

Carbon monoxide, CO, reacts with hydrogen gas, H_2, to produce propane, C_3H_8 and water according to the equation: $CO_{(g)} + H_{2(g)} \rightarrow C_3H_{8(g)} + H_2O_{(l)}$. What volume in L of C_3H_8 at 32.0 ◦C and 2.50 atm is produced by the reaction of 5.50 g CO gas?

Given: mass of CO = 5.50 g, P = 2.50 atm, and T = 32.0 ◦C + 273 = 305 K
Required: Volume of C3H8 in L
Solution: STEP 1: Balance the equation.

$$3CO_{(g)} + 7H_{2(g)} \rightarrow C_3H_{8(g)} + 3H_2O_{(l)}$$

STEP 2: Get the necessary mole-mole relationship from the balanced equation.

3 moles CO = 1 mole C_3H_8

STEP 3: Determine the mass of C_3H_8 produced from 5.50 g of CO.

5.50 g CO x (1 mole CO)/(28.01 g) x (1 mole C_3H_8)/(3 moles CO)

Note: Cancel the same units between numerators and denominators.

= (5.50 mole C_3H_8)/84.03

n = 0.0655 mole C_3H_8

STEP 4: Use the Ideal Gas law equation to solve for the Volume, V.

PV = nRT V = nRT/P

R = the gas constant, 0.0821 L-atm/ mol K

STEP 5: Substitute the given to the formula.

V = (n RT)/P **Note:** Cancel the same unit

= ((0.0655 mole)(0.0821 (L-atm)/(mol K) (305 K)))/(2.50 atm)

= (1640 L)/2.50

V = 0.656 L

What's More

Directions: Read and follow the steps in solving the problem below.

Ammonium Nitrate NH_4NO_3, decomposes to produce nitrogen gas, N_2, liquid water H_2O, and oxygen gas, O_2. The equation for this reaction is: $NH_4NO_{3(s)} \rightarrow N_{2(g)} + H_2O_{(l)} + O_{2(g)}$. Determine the volume in ml of O_2 produced at 85^{0}C and 2.0 atm when 8.50 g NH_4NO_3 is decomposed.

1. Balance the equation.
2. Find the mole-mole relationships from the balanced equation.
3. Determine the mass of O_2 produced from 8.50 g of NH_4NO_3.
4. Use the Ideal Gas Law equation.
5. Substitute the given to the formula.

What I Have Learned

Directions: Complete the paragraph by filling in the blanks with the correct word. Choose your answer from the box below.

Behavior	gas laws	products	properties	reactants

When we perform experiments that involve gaseous _____ and _____, the knowledge of the _____ and properties of gases become important. In such reactions, the basic principle of _____ applies in addition to different _____.

What I Can Do

Directions: Read and answer the question below.

Based on the reaction, is it safe to use muriatic acid to clean limestone tiles?

_____

_____

_____

Assessment

Directions: Solve the problem below. Use another sheet of paper for your answer.

Limestone rocks are composed mainly of calcium carbonate $CaCO_3$ that reacts easily with an acid, releasing carbon dioxide gas: $CaCO_{3(s)} + 2HCl_{(aq)} \rightarrow CaCl_{2(aq)} + CO_{2(g)} + H_2O_{(l)}$. How much CO_2 in L will be produced by 350 g of $CaCO_3$ if it completely reacted with HCl at 25^0 C and 1 atm?

Additional Activities

Directions: Stoichiometry of gases is considered useful in many walks of life. List down 5 uses of gas stoichiometry in various daily life activities.

1. _____
2. _____
3. _____
4. _____
5. _____

Lesson 12 Relate the Rate of Gas Effusion with Molar Mass

According to Kinetic Molecular Theory, gaseous particles are in a constant state of motion, moving at random speeds and in many different directions. Because of their kinetic energy at temperatures above absolute zero, all particles undergo diffusion.

At the end of this unit, you are expected to:

1. Relate the rate of gas effusion with molar mass. **STEM_GC11KMTIj-50**

What's In

Thomas Graham, Scottish chemist (1805-1869) formulated the law of effusion in 1848. It states that the rate of effusion or diffusion of gas is inversely proportional to the square root of its molar mass. This means that the heavier the molecule, the slower is its motion. The mathematical expression of Graham Law of Effusion or Diffusion can be written as:

$$\text{Rate of Effusion} \propto \frac{1}{\sqrt{molar\ mass}}$$

The term effusion and diffusion are often interchangeably, but they are not equal. **Effusion** is the movement of gas particles through a tiny hole. **Diffusion** is the process by which a gas spreads or mixes with another gas. The ability of gases to diffuse enables us to recognize odors and scents.

What's New

If you have ever been in a room when a piping hot pizza was delivered, you have been made aware of the fact that gaseous molecules can quickly spread throughout a room, as evidenced by the pleasant aroma that soon reaches your nose. Although gaseous molecules travel at tremendous speeds (hundreds of meters per

second), they collide with other gaseous molecules and travel in many different directions before reaching the desired target. At room temperature, a gaseous molecule will experience billions of collisions per second.

What Is It

Graham's law predicts that the lighter gas particles will move faster than heavier ones. As the kinetic molecular theory postulates, the average kinetic energy of a gas is proportional to its absolute temperature. The Kinetic Molecular Theory of gases clearly explains why gas molecules easily diffuse. The wide distances between gas molecules and their low densities enable them to move faster.

The speed of the gas particles and their molar masses gives the Graham's law equation.

$$\frac{r1}{r2} = \sqrt{\frac{MM2}{MM1}}$$ where: r = rate MM = molar mass

Note: Graham's law equation can be modified if there is a time given.

$$\frac{t2}{t1} = \sqrt{\frac{MM2}{MM1}}$$ where: t = time MM = molar mass

Sample Problem 1

What is the molar mass of gas A if it diffuses 0.8 times as rapid as O_2?

Given: r1 = 1 MM1 = O_2 = 16g/mole x 2 = 32 g/mole
r2 = 0.8 MM2 = gas A **Note:** 16 g/mole is the mass of O_2
Required: Molar mass of gas A/MM2
Solution: STEP 1: Write the formula and derive the formula to be used.

$$\frac{r1}{r2} = \sqrt{\frac{MM2}{MM1}} \qquad \sqrt{MM2} = \frac{(r1)\sqrt{MM1}}{r2}$$

STEP 2: Substitute the given to the formula.

$$\sqrt{gas\ A} = \frac{(1)\ \sqrt{32g/mole}}{0.8}$$

$$\sqrt{gas\ A} = \frac{\sqrt{(32g/mole)}}{0.8}$$

$$(\ gas\ A\ )^2 = \frac{5.66\ g/mole}{0.8}$$

$$= (7.075\ g/mole)^2$$

gas A = 50.01 g/mole

Sample Problem 2

Calculate the ratio of rate of effusion of methane gas, $CH_{4(g)}$ to $He_{(g)}$ at constant temperature and pressure. The molar mass for $CH_{4(g)}$ and $He_{(g)}$ are 16.0 g/mole and 4.0 g/mole respectively.

Given: MM1 = CH_4 = 16.0 g/mole, MM2 = He = 4 g/mole
Required: Ratio of rate of effusion of $CH_{4(g)}$ and $He_{(g)}$
Solution: Step 1: Write the formula to be used.

$$\frac{r1}{r2} = \sqrt{\frac{MM2}{MM1}}$$

Step 2: Substitute the given to the formula.

$$\frac{rCH_4}{r\ He} = \sqrt{\frac{4\ g/mole}{16\ g/mole}}$$ **Note:** Cancel units.

$$\frac{rCH_4}{rHe} = \frac{2}{4} = \frac{1}{2}$$ **Note:** Simplify the answer.

The ratio of the two gases is 1:2.

What's More

Directions: Analyze and solve the problem below.

It took 6.3 minutes for 25 ml of Argon to reach the end of an effusion tube with a porous barrier. How long will it take for 25 ml of $CHCl_3$ to effuse through the tube under the same condition?

Note: since the problem is asking for the distance, Graham's law of effusion can be modified to

$$\frac{t2}{t1} = \sqrt{\frac{MM2}{MM1}}$$

What I Have Learned

Directions: Answer the question to the best of your ability.

Why is the movement of a gas inversely proportional to its molar mass?

_____

_____

What I Can Do

Directions: Read and answer the question below.

How does the Graham's law explain why we smell odors? Explain your answer.

_____

_____

Assessment

Directions: A. Analyze and solve the problems. Use another sheet of paper for your answer.

1. Oxygen gas, O_2, is observed to diffuse through a capillary for 18 seconds. How long would it take for methane, CH4, gas to diffuse through the same orifice under identical conditions of temperature and pressure?
2. Determine the ratio of the distances through which CO_2 and C_2H_2 will effuse at constant temperature and pressure.

B. Write *TRUE* if the statement is correct and *FALSE* if incorrect.

_____ 1. LPG and highly flammable gases should be stored in heavy tanks and must be placed in cool places.

_____ 2. Exposing them to sunlight should not be avoided.

_____ 3. Gas molecules move faster when heated. Increase in the movements of gas molecules increases the pressure they exert on the walls of their container which may result in an explosion.

_____ 4. The LPG tanks should be tightly closed when not in used to avoid accidents due to leakage of the flammable methane.

_____ 5. Empty containers of aerosol sprays must be hammered or punctured to avoid untoward occurrences.

Additional Activities

Directions: Read and answer the question below.

Why is the movement of a gas inversely proportional to its molar mass?

_____

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For inquiries or feedback, please write or call:

Department of Education - Valenzuela
Pio Valenzuela Street, Marulas
Valenzuela City
Telefax: 8-292-4340
Email Address: sdovalenzuela@deped.gov.ph